Smart Intrusion Detection System

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Industrial Internet of Things Project Report

GitHub Repository

The complete project source code, simulation files, and documentation are available at:

https://github.com/sas-1421/24BEC1450-Sashank-Muvvala-Smart-Intrusion-Detection-System-IIOT-project-

Figure 1: QR Code for GitHub Repository

Objective/Aim

To design and simulate a Smart Intrusion Detection System using an ESP32 microcontroller, PIR motion sensor, and ThingSpeak cloud platform for detecting unauthorized human movement and providing real-time monitoring through IoT-based communication.

Problem Statement

Security and surveillance are critical requirements in homes, offices, laboratories, warehouses, and other restricted areas. Traditional monitoring systems often rely on continuous human supervision, making them inefficient and prone to delayed responses during unauthorized access. There is a need for an automated system that can continuously monitor a protected area, detect human movement, and provide immediate alerts to security personnel.

The objective of this project is to develop a Smart Intrusion Detection System using an ESP32 microcontroller and a PIR motion sensor. The system continuously detects human movement in a restricted area and identifies potential intrusion events. Upon detecting motion, the ESP32 activates local alert devices such as an LED and buzzer while simultaneously transmitting the event data to the ThingSpeak cloud platform through a Wi-Fi connection.

The cloud-based monitoring capability enables real-time visualization of intrusion events and allows remote supervision of the protected area. The proposed solution aims to provide a low-cost, reliable, and scalable security system capable of enhancing safety through automated detection and instant notification of unauthorized access.

Scope of the Solution

The proposed Smart Intrusion Detection System is designed to monitor restricted areas and detect unauthorized human movement using a PIR motion sensor. The system provides local alerts through an LED and buzzer while simultaneously transmitting intrusion data to the ThingSpeak cloud platform for remote monitoring.

The solution is suitable for applications such as homes, offices, laboratories, warehouses, and industrial facilities. The project focuses on real-time motion detection, cloud-based monitoring, and automated alert generation using IoT technology.

Future enhancements may include mobile notifications, camera integration, GSM-based alerts, event logging, and AI-assisted threat analysis.

Introduction

Security monitoring systems play a crucial role in protecting homes, offices, laboratories, warehouses, and industrial facilities from unauthorized access. Traditional surveillance systems often require continuous human supervision and may fail to provide immediate notifications when intrusions occur.

The Smart Intrusion Detection System presented in this project utilizes Internet of Things (IoT) technology to automate intrusion monitoring and alert generation. A Passive Infrared (PIR) sensor continuously monitors a restricted area for human movement. Whenever motion is detected, the sensor sends a signal to the ESP32 microcontroller.

The ESP32 processes the sensor data and activates local alert devices such as an LED and buzzer. Simultaneously, the controller connects to the internet through Wi-Fi and transmits intrusion information to the ThingSpeak cloud platform. The transmitted data can be visualized remotely, enabling real-time monitoring and faster response to security threats.

The proposed system provides a low-cost, scalable, and reliable solution for intrusion detection while demonstrating the practical application of IoT technology in security systems.

Components Required

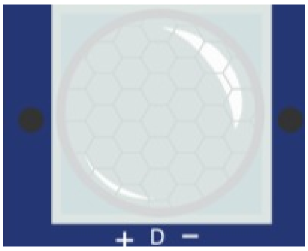
The following hardware and software components were used to develop the Smart Intrusion Detection System.

ESP32 DevKit V1

The ESP32 is the main controller of the system. It processes the PIR sensor data, controls the LED and buzzer alerts, and communicates with the ThingSpeak cloud platform through its built-in Wi-Fi module.

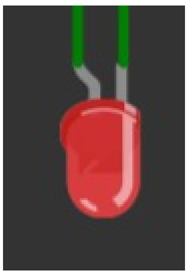
PIR Motion Sensor

The Passive Infrared (PIR) sensor detects human movement by sensing changes in infrared radiation. It acts as the primary sensing element of the intrusion detection system.



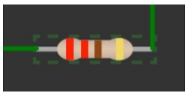
Red LED

The LED provides a visual indication whenever motion is detected. It helps users identify intrusion events immediately.



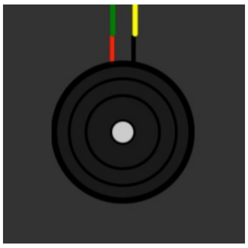
220Ω Resistor

The resistor is connected in series with the LED to limit current flow and protect the LED from damage.



Active Buzzer

The buzzer generates an audible alert whenever motion is detected, providing immediate warning of unauthorized access.



ThingSpeak Cloud Platform

ThingSpeak is an IoT cloud platform used to store, visualize, and monitor sensor data in real time. The ESP32 uploads motion status data to ThingSpeak through HTTP requests.

Wokwi Simulator

Wokwi is an online simulation platform used to design, test, and validate embedded systems without requiring physical hardware.

Methodology / Implementation

Motion Detection using PIR Sensor

The PIR sensor continuously monitors infrared radiation emitted by nearby objects. Human movement causes variations in infrared energy which are detected by the sensor. The sensor output pin is connected to GPIO13 of the ESP32.

Intrusion Processing using ESP32

The ESP32 acts as the central processing unit of the system. It continuously reads the output of the PIR sensor and determines whether motion has occurred within the monitored area.

Alert Generation

Whenever motion is detected, the ESP32 activates a red LED connected to GPIO2 and generates an audible warning using a buzzer connected to GPIO4. These alerts provide immediate local indication of intrusion.

Cloud Communication

The ESP32 establishes a Wi-Fi connection and communicates with the ThingSpeak cloud platform. Motion events are transmitted through HTTP requests, allowing remote monitoring through the cloud dashboard.

Real-Time Monitoring

The system continuously repeats the detection process and updates the cloud platform whenever the sensor state changes. This ensures real-time monitoring and rapid notification of intrusion events.

Code

ESP32 and ThingSpeak Integration

```
#include <WiFi.h>
#include <HTTPClient.h>

// WiFi credentials
const char* ssid     = "Wokwi-GUEST";
const char* password = "";

// ThingSpeak
const char* apiKey   = "API KEY";
const char* server   = "http://api.thingspeak.com/update";

const int PIR_PIN = 13;
const int LED_PIN = 2;
const int BUZZER_PIN = 4;

void setup() {
Serial.begin(115200);
pinMode(PIR_PIN, INPUT);
pinMode(LED_PIN, OUTPUT);

WiFi.begin(ssid, password);

Serial.print("Connecting to WiFi");

while (WiFi.status() != WL_CONNECTED) {
delay(500);
Serial.print(".");
}

Serial.println("\nWiFi Connected!");
delay(2000);
}

void loop() {

int motion = digitalRead(PIR_PIN);

if (motion == HIGH) {
Serial.println("ALERT: Motion Detected!");
digitalWrite(LED_PIN, HIGH);


sendToThingSpeak(1);

delay(15000);
```

```
}
else {


Serial.println("No Motion");
digitalWrite(LED_PIN, LOW);

sendToThingSpeak(0);

delay(15000);


}
}

void sendToThingSpeak(int value) {

if (WiFi.status() == WL_CONNECTED) {


HTTPClient http;

String url = String(server)
           + "?api_key="
           + apiKey
           + "&field1="
           + value;

http.begin(url);

int httpCode = http.GET();

if (httpCode > 0) {
  Serial.println("ThingSpeak Updated");
}
else {
  Serial.println("Transmission Failed");
}

http.end();


}
}
```

Flowchart/Block Diagram

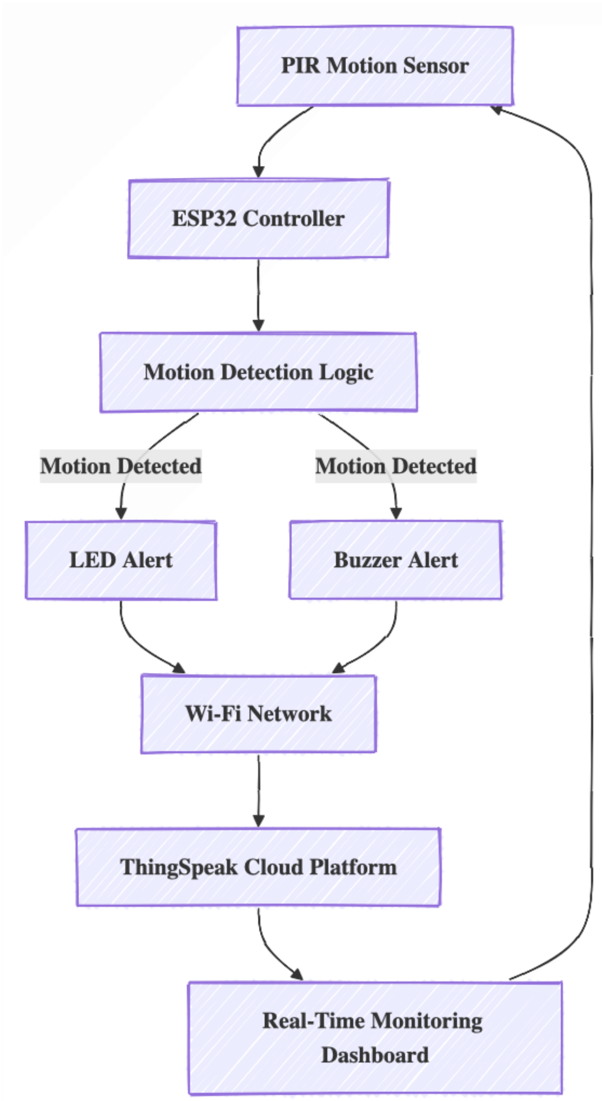


Figure 2: Flowchart / Block Diagram

Circuit Diagram

The Smart Intrusion Detection System was implemented using an ESP32, PIR motion sensor, LED indicator, resistor, and buzzer. The PIR sensor output is connected to GPIO13 of the ESP32. The LED is connected to GPIO2 through a 220Ω resistor and the buzzer is connected to GPIO4.

The ESP32 receives motion information from the PIR sensor and updates the ThingSpeak cloud platform through Wi-Fi communication.

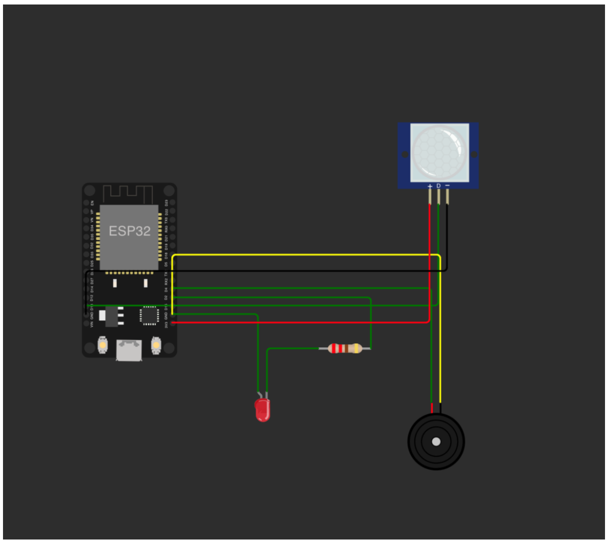


Figure 3: Wokwi Circuit Diagram of Smart Intrusion Detection System

Simulation Output

Before deployment, the complete system was tested using the Wokwi simulation environment. The simulation validated the operation of the PIR sensor, ESP32 controller, LED indicator, buzzer, Wi-Fi communication, and cloud integration.

During simulation, motion events generated by the PIR sensor were successfully detected by the ESP32. The LED and buzzer responded immediately to intrusion events. The ESP32 transmitted motion data to ThingSpeak using HTTP requests.

The simulation verified:

- Accurate motion detection
- Correct operation of ESP32 processing logic
- LED alert generation
- Buzzer alert generation
- Successful Wi-Fi communication
- Real-time cloud monitoring
- Reliable ThingSpeak updates

The ThingSpeak graph displayed values of 1 during motion detection and 0 during normal conditions, confirming successful cloud communication and monitoring.

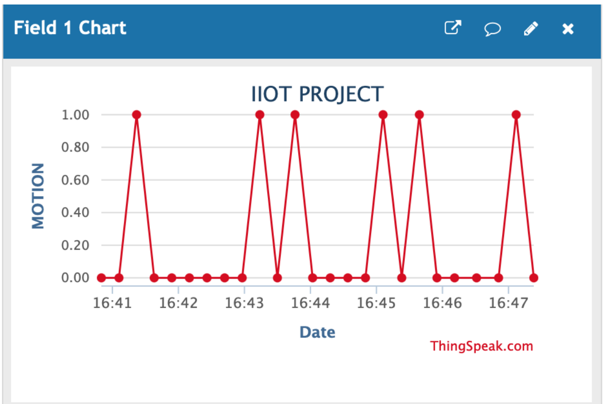


Figure 4: ThingSpeak Cloud Monitoring Output

Demonstration Video

A demonstration video showing the operation of the Smart Intrusion Detection System, including motion detection, LED indication, cloud updates, and ThingSpeak monitoring, has been uploaded to the GitHub repository

Results

The Smart Intrusion Detection System successfully detected motion in the monitored area and generated real-time alerts.

When motion was detected:

- The PIR sensor generated a HIGH output signal.
- The ESP32 detected the intrusion event.
- The LED turned ON.
- The buzzer generated an audible alert.
- Motion data was uploaded to ThingSpeak.
- The cloud dashboard displayed the intrusion event.

The generated cloud graph confirmed successful communication between the ESP32 and ThingSpeak platform.

Conclusion

The Smart Intrusion Detection System successfully demonstrates the application of Internet of Things technology in modern security systems. By integrating a PIR motion sensor, ESP32 microcontroller, LED indicator, buzzer, and ThingSpeak cloud platform, the system provides an effective solution for detecting unauthorized access and generating real-time alerts.

The project validated reliable motion detection, local alert generation, wireless communication, and cloud-based monitoring. The use of ThingSpeak enabled remote visualization of intrusion events and enhanced the overall effectiveness of the security system.

Future improvements may include mobile notifications, GSM alerts, image capture, cloud data logging, and AI-based threat analysis. Overall, the proposed solution provides a scalable, low-cost, and efficient approach for intrusion detection applications.

References

1. ESP32 Documentation: https://documentation.espressif.com/en/home
2. ThingSpeak Documentation: https://thingspeak.com
3. Wokwi Simulator: https://wokwi.com
4. Arduino Documentation: https://www.arduino.cc
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6. MathWorks ThingSpeak Platform: https://thingspeak.mathworks.com
7. L&T EduTech IIOT STUDY MODULES